

# Gangmin Yu

Mechanical Engineering · Undergraduate Researcher

Kyonggi University · RVLab

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## RESEARCH INTERESTS

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Autonomous driving and navigation, vehicle dynamics and control, radar/sensor perception, robot simulation with Isaac Sim, robot mechanical design, embedded and mechatronic systems.

## EDUCATION

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**Kyonggi University**, Suwon, Republic of Korea

Mar. 2021 – Present

B.S. in Mechanical Engineering

Undergraduate Researcher, RVLab (Advisor: Prof. Minseong Choi)

**GPA:** 4.27 / 4.5   **Major GPA:** 4.42 / 4.5

**English:** TOEIC Speaking IH (150)

## PUBLICATIONS

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### International Journals (*On going*)

Gangmin Yu\*, Minjun Seong\*, and Minseong Choi. “4D Radar Ego Velocity Estimation Network for Autonomous Driving” *IEEE Sensors*, 2026.

### Domestic Conferences

이충헌, 유강민, 박영상, 최민성, 조우성. “Swerve 구동 휴머노이드를 위한 Holonomic 실내 자율 탐색 알고리즘” *제어·로봇·시스템학회(ICROS) 학술대회*, 2026.

김도윤, 강민수, 유강민, 서형태, 박영상, 최민성. “IsaacSim 기반 휴머노이드 ACT 학습제어를 위한 VR 원격조작 및 Jitter 저감 기법” *제어·로봇·시스템학회(ICROS) 학술대회*, 2026.

성민준, 유강민, 김준규, 한승호, 양승훈, 최민성. “도로 노면 의미론적 분할 기법의 소수 클래스 불균형 해소를 위한 원근 변환 기반 데이터 증강 기법” *제어·로봇·시스템학회(ICROS) 학술대회*, 2026.

## RESEARCH & PROJECTS

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### ONGOING

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#### 4D Radar Ego-Velocity Estimation

RVLab, Kyonggi University

- Estimating ego-vehicle velocity directly from 4D radar measurements toward robust self-motion estimation for autonomous driving.

#### Manipulation & Navigation for Humanoids in Domestic Environments

RVLab

- Applied physics-based digital-twin simulation and MetaQuest 3 teleoperation in Isaac Sim.
- Implemented autonomous driving and navigation using Nav2 on a generated 2D LiDAR map.

### PAST

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#### 3D Simulation Environments & Physics Modeling for Daily-Life Scenes

KIMM (*commissioned*)

*funded by Korea Institute of Machinery & Materials (KIMM)*

- Built physically simulated indoor environments in NVIDIA Isaac Sim (PhysX) for realistic robot interaction.
- Authored rigid-body and collider models from .obj meshes; set per-object mass, inertia, friction, restitution.

- Defined articulated structures (hinge/prismatic joints) for doors and drawers with motion limits and dynamics.

#### **Sensor Mount Design for ROBOTIS AI Worker**

*RVLab · Humanoid Hardware*

- Designed a torso-mounted bracket to integrate an additional 3D LiDAR.
- Designed an angle-adjustable gripper camera mount to improve field of view during grasping.

#### **Autonomous Vehicle Build**

*Autonomous Driving Club*

- As club leader, oversaw the entire vehicle build process, designing the hardware/embedded system (actuators, sensors, MCU communication).
- Implemented steering control (Pure Pursuit) and speed control (PID).

#### **High-Voltage Shutdown Circuit**

*Self-Built Car Club · Formula (EV)*

- Designed a safety shutdown circuit for 120 V high voltage, cutting power on BMS/IMD/interlock faults or sudden-acceleration detection.
- Designed and fabricated the PCB (KiCad) and applied it to the vehicle.

#### **Braking System Design**

*Self-Built Car Club · Formula & Baja*

- Computed braking force from vehicle kinematics and designed the brake disc accordingly.
- Used CAE to optimize the design and verify structural safety.

## **AWARDS**

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**Excellence Award** — Self-Built Autonomous Vehicle Division (Leader)

*Nov. 2025*

University Creative Mobility Competition · Korea Transportation Safety Authority (KOTSA)

**Acceleration Award** — Baja Student (Team Member)

*Nov. 2021*

International Self-Built Car Competition · Yeungnam University

## **TECHNICAL SKILLS**

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**Simulation / Robotics:** NVIDIA Isaac Sim, PhysX, ROS2, AMCL / Navigation

**Control / Embedded:** Embedded C, PID, Pure Pursuit, MCU & Sensors

**Design / CAE:** KiCad (Schematic / PCB), AutoCAD, CATIA, Ansys

**Domain:** Autonomous Driving, Vehicle Dynamics, Radar Perception, Mechatronics